

Open-Set Bat Face Recognition: Progress Report

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Goal. Build a non-invasive system that identifies individual fruit bats from facial images, supporting long-term behavioural studies without physical tagging. The system handles *open-set* scenarios: bats seen at test time were never observed during training.

Current status. A three-model comparison (Siamese pair network, ArcFace, AdaFace) has been completed across two species (rousettus, mauritius) and three background conditions, using a rigorous identity-disjoint evaluation protocol. A fully automated training and reporting pipeline is operational.

1 Key Finding: Evaluation Protocol Matters

A common pitfall in animal re-identification is *identity leakage*: splitting images randomly instead of splitting identities means the same bat appears in both training and test sets, inflating results. Table 1 demonstrates the scale of this effect.

Table 1: Same model, same data — different splitting protocol.

Protocol	ROC-AUC	F ₁	Recall
Random-pair split (leaky)	0.982	0.987	0.986
Identity-disjoint split (correct)	0.805	0.742	0.734

Takeaway: all results in this report use identity-disjoint splits exclusively.

2 Model Comparison

2.1 Rousettus — 12 individuals, 1 059 images (6/3/3 id split)

Table 2: Test-set results on rousettus (random background, 3 unseen test identities). Pair-family uses verification metrics (ROC-AUC, F₁); embedding-family also reports identification metrics.

Model	HPs	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	tuned	0.809	0.747	—	—	0.017
ArcFace	tuned	0.541	—	0.344	0.634	0.000
AdaFace	tuned	0.507	—	0.286	0.609	0.004
ArcFace	default	0.500	—	0.326	0.633	0.000

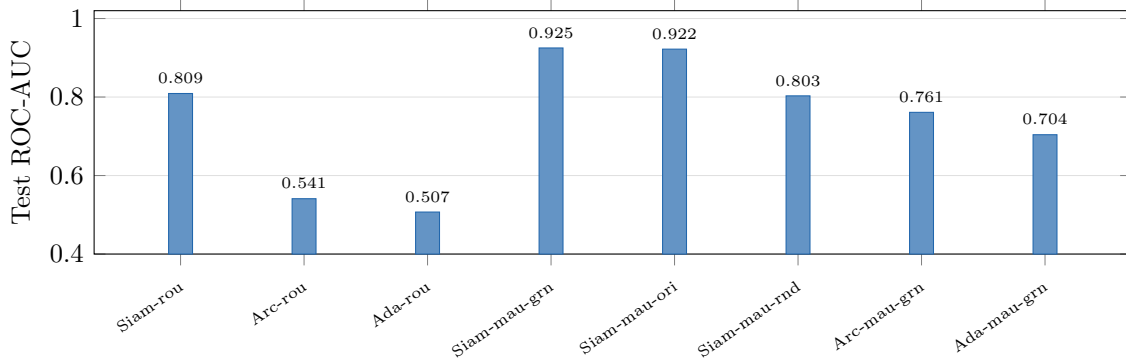
Note on ArcFace default: paper-scale hyperparameters (SGD $\eta=0.1$, margin 0.5, scale 64) destroy the ImageNet pretrained backbone on 486 training images — both train accuracy and validation ROC-AUC collapse to chance. Tuning to Adam $\eta=10^{-3}$, margin 0.2, scale 30 recovers partial performance but still falls far short of the Siamese network.

2.2 Mauritius — 11 individuals, 1 119 images (5/3/3 id split)

Table 3: Test-set results on mauritius across background conditions.

Model	Background	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	green	0.925	0.864	—	—	0.083
Siamese	original	0.922	0.872	—	—	0.041
Siamese	random	0.803	0.759	—	—	0.001
ArcFace	green	0.761	—	0.706	0.841	0.044
AdaFace	green	0.704	—	0.566	0.778	0.022
ArcFace	random	0.526	—	0.362	0.618	0.000

3 ROC-AUC Summary (bar chart)



x-axis key: Siam = Siamese, Arc = ArcFace, Ada = AdaFace; rou = rousettus (random bg); mau = mauritius with grn/ori/rnd background.

4 Conclusions & Next Steps

- **Pair-based (Siamese) outperforms embedding models at this scale.** With ≤ 12 training identities, angular-margin losses (ArcFace/AdaFace) learn closed-set geometry that does not generalise to unseen test identities (train accuracy $\approx 100\%$, but test ROC-AUC near chance). The Siamese pairwise objective generalises more naturally.
- **Background matters.** Green-screen or original backgrounds outperform random compositing by ~ 12 ROC-AUC points on mauritius, suggesting controlled data collection is a high-leverage improvement.
- **Immediate next step: more identities.** The small-identity wall (≤ 12) is the primary bottleneck. Expanding the rousettus manifest to 30+ individuals is the highest-priority action to determine whether ArcFace/AdaFace can close the gap.
- **Reproducibility.** All experiments are tracked in MLflow and reproducible with a single command:
`uv run bat-cli train -experiment <name>.`